

Turing Instrument Calibration and Validation Program

Version 1.0: Experimental Protocol

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Supplementary protocol to the preprint “Computing Machinery and Understanding: From the Imitation Game to the Turing” (v1.7). The preprint defines the Turing unit and its falsification criteria; this document specifies the ordered experimental program that determines whether the unit is a scientifically useful measure. This protocol is versioned independently of the paper.

1 Purpose

This program is designed to determine whether the **Turing** is a scientifically useful, reproducible measure of generalized explanatory compression rather than merely a mathematically defined scoring function.

The experimental program must answer, in order:

1. Does theoretical predictive description length correspond to actual lossless compression?
2. Does the metric reward genuine stochastic structure while rejecting memorization and overfitting?
3. Does additional model structure receive credit only when it earns its complexity?
4. Can the metric distinguish qualitatively different generating mechanisms?
5. Can it reward useful unification across multiple phenomena?
6. Does an explanation remain useful under interventions and distributional changes?
7. Can explanatory models help select better experiments?
8. Are results reproducible under implementation replay and fresh stochastic worlds?
9. Can the associated physical energy accounting be independently closed?
10. Can equivalent explanations be distinguished by explanatory efficiency in Turings per joule?
11. Are scientific conclusions stable under reasonable alternative measurement profiles?
12. Can an independent laboratory reproduce the conclusions?

The program is deliberately structured so that each major hypothesis can fail. Large positive Turing values are not sufficient evidence. The important question is whether the instrument makes correct preregistered discriminations in cases where it is possible for the metric to be wrong.

2 Core Experimental Principle

The experiments must be executed as sequential qualification gates. The canonical sequence is:

```
CAL-0
Freeze measurement apparatus
->
EXP-001
Connect theoretical description length
to actual lossless compression
->
EXP-002A
Reject false structure in pure randomness
->
EXP-002B
Test when additional structure earns its cost
->
EXP-002C
Distinguish qualitatively different mechanisms
->
EXP-002D
Test explanatory unification
->
Interventional validation
Test frozen laws outside the observational regime
->
EXP-003
Select informative experiments
->
H3
Reproducibility
->
H4
Physical energy accounting
->
H5
Turing yield
->
Sensitivity analysis
->
Independent laboratory replication
```

A downstream success does not repair an upstream failure. If EXP-001 fails, later experiments may continue for debugging purposes, but they cannot support claims that depend upon a validated Turing measurement.

3 Measurement Quantity

For candidate model or program M , baseline B , and sealed held-out data D , calculate:

$$T(M; B, D) = [L(B) + L(D|B)] - [L(M) + L(D|M)].$$

The measurement must account for both $L(M)$, the description cost of the explanation itself, and $L(D|M)$, the description length of observations conditional on that explanation.

Predictive codelength is initially calculated as:

$$L_{\text{ideal}}(D|M) = - \sum_t \log_2 p_M(x_t | x_{<t}).$$

EXP-001 then determines whether this idealized quantity corresponds closely enough to an actual losslessly encoded artifact to support physical compression claims.

4 CAL-0: Freeze the Measurement Apparatus

4.1 Objective

Before evaluating any scientific claim, create and freeze a canonical measurement profile:

Turing-profile-v1.0

The profile may not be modified during the first validation series. A change to the profile creates a new profile version. Results under different profiles must not be silently treated as numerically interchangeable.

4.2 Required Freeze Order

The required sequence is:

```
experiment designed
->
candidate families specified
->
statistical simulations performed
->
sample size selected
->
pass/fail rules preregistered
->
measurement profile frozen
->
candidate development completed
->
candidate artifact frozen and hashed
->
sealed evaluation data generated/released
->
evaluation performed
->
results retained whether successful or unsuccessful
```

Candidate development must end before the evaluator obtains sealed test observations. A cryptographic commitment alone does not establish blinding if the candidate has already had access to the hidden data.

5 Turing-profile-v1.0

The frozen profile must contain at minimum the following fields.

5.1 Identity

profile_name
profile_version
profile_digest
creation_timestamp
protocol_version

5.2 Experimental Definition

task_definition
hypothesis_definition
primary_endpoint
secondary_endpoints
pass_fail_rule
stopping_rule

5.3 Baseline

baseline_family
baseline_encoding
baseline_parameterization
baseline_description_cost

5.4 Candidate Encoding

candidate_representation
candidate_serialization
shared_background
model_code_method
parameter_precision
dependency_cost
metadata_cost
lookup_table_cost
external_artifact_cost

Particular attention must be paid to $L(M)$. The experiment must define exactly what information is considered free shared background and what information must be charged to the candidate. No model may hide predictive information in metadata, filenames, headers, preloaded tables, dependencies, or uncharged support files.

5.5 Data

data_generator
development_split
calibration_split
sealed_test_split
replication_split
trajectory_count
trajectory_length
sampling_interval
seed_commitment

5.6 Blinding

holdout_commitment
candidate_environment
evaluator_environment
access_control
test_release_process
candidate_freeze_process

5.7 Probabilities

probability_representation
quantizer
normalization_rule
zero_probability_policy
numeric_precision

5.8 Coding

ideal_codelength_method
reference_coder_A
reference_coder_B
termination_convention
header_accounting
side_information_accounting

5.9 Statistics

uncertainty_method
confidence_level
block_length
bootstrap_method
correlation_handling
number_of_replicates

5.10 Physical Measurement

energy_instrument

sampling_frequency
idle_baseline_method
attribution_method
sensor_uncertainty
time_alignment_method

5.11 Reproducibility

software_versions
compiler_versions
hardware_identity
runtime_digest
candidate_digest
dataset_digest
coder_digest
scorer_digest

5.12 Sensitivity

alternative_baselines
alternative_model_codes
alternative_quantization
alternative_reference_coder
alternative_uncertainty_model

6 Sample-Size Determination

The manuscript does not prescribe a fixed sample size. Therefore the experiment must not choose one arbitrarily. Before preregistration:

1. simulate the proposed hidden processes;
2. simulate the expected range of candidate effect sizes;
3. apply the proposed uncertainty procedure;
4. determine how many independent worlds or trajectories are required to distinguish effects considered scientifically meaningful;
5. freeze that sample size before final sealed-data generation.

The simulation itself must be archived. The sample-size calculation must not be rerun with revised assumptions after sealed outcomes are known unless the original experiment is formally declared inconclusive and a new experiment is preregistered.

7 Experimental Roles

At minimum, separate the following roles.

7.1 Candidate Team

May access:

- public protocol
- development observations
- development simulators where permitted
- public candidate APIs

Must not access:

- sealed seeds
- sealed trajectories
- hidden generator parameters
- evaluator secrets
- future intervention results

7.2 Evaluator

Controls:

- sealed observations
- sealed parameters
- reference coding
- scoring
- experiment execution

7.3 Verification Team

Must independently reconstruct:

- model cost
- ideal codelength
- actual coded length
- Turing value
- uncertainty

without depending on the production scoring path.

7.4 Replication Team

Uses independent seeds and preferably independent implementations.

7.5 External Replication Team

Ultimately generates its own hidden data and implements the measurement protocol independently.

8 EXP-001: Compression Bridge

8.1 Scientific Question

Does the theoretical predictive codelength used by the Turing correspond to actual physical lossless compression? The experiment must close:

```
probability prediction
->
-log2 probability
->
entropy coder
->
real encoded bitstream
->
real compressed size
```

The current theoretical score alone is insufficient to establish this connection.

8.2 Candidate Predictors

Freeze at least:

ID	Candidate	Purpose
B0	uniform	minimal reference
B1	empirical frequency	simple statistical baseline
B2	small Markov	short-range structural baseline
B3	fixed heuristic	domain-specific baseline
M1	explanatory candidate	primary model
Mmem	memorizer	adversarial complexity control

9 Probability Artifact

Every predictor must first produce an immutable probability stream. Each prediction record should include:

```
observation_index
context_digest
alphabet
raw_probability_vector
quantized_probability_vector
normalization_receipt
```

Freeze and hash the probability stream. Both coders must consume exactly this stream. They must not independently invoke the model. This separates model error from coder error.

10 Dual Reference Coders

Use two independently implemented coder families. Recommended:

Coder A:
arithmetic or range coder

Coder B:
ANS or rANS

For each candidate require:

$$\text{decode}(\text{encode}(D)) = D.$$

The decoded sequence must exactly equal the sealed sequence. The two coders need not produce identical binary streams. Their conclusions should agree within the preregistered coding-overhead envelope.

11 EXP-001 Measurements

For every candidate compute $L(M_i)$, $L_{\text{ideal}}(D|M_i)$, $L_{\text{coded-A}}(D|M_i)$, and $L_{\text{coded-B}}(D|M_i)$. Then calculate:

T_ideal
T_coder_A
T_coder_B

and:

coder_A_overhead
coder_B_overhead

12 EXP-001 Adversarial Controls

Deliberately test:

corrupted bitstream
truncated bitstream
incorrect profile digest
incorrect model digest
incorrect test-data digest
invalid probability normalization
zero-probability actual event
hidden metadata
candidate/test overlap
unaccounted headers
unaccounted side information
memorized data disguised as program state

Every invalid condition must fail closed.

13 EXP-001 Acceptance Criteria

Require:

both decoders reproduce the sealed data exactly

candidate hashes predate sealed-data release

both coders consume the same probability artifact

actual coded lengths agree with ideal lengths
within preregistered coder overhead

scientifically relevant model ordering remains stable

positive candidate T survives real coding
if preregistered to do so

memorizer does not receive unjustified positive generalized T

independent scorer reproduces the published result

new-seed replication preserves the qualitative conclusion

Gate:

EXP_001_COMPRESSION_BRIDGE = PASS

or:

EXP_001_COMPRESSION_BRIDGE = FAIL

A failed result must be retained.

14 EXP-002A: Randomness Without Invented Structure

14.1 Hidden Process

Generate:

$$dX_t = \sqrt{2D} dW_t.$$

The evaluator privately samples:

D
initial condition
trajectory
seed

The scientific question is whether the metric discovers centered stochastic diffusion without inventing deterministic patterns from random fluctuations.

15 Candidate Families

Compare:

weak baseline

simple diffusion

diffusion + drift

mean-reverting model

polynomial/curve fit

high-capacity flexible predictor

memorizer

The correct structural family is deliberately simple. The experiment must provide attractive opportunities for overfitting.

16 EXP-002A False-Structure Challenge

Include development trajectories with accidental apparent:

trend

curvature

local periodicity

reversal

generated purely by noise. Do not remove these runs. They provide the most informative test. A polynomial or flexible predictor may achieve excellent development fit. Its held-out performance must determine whether that structure actually earns explanatory credit.

17 EXP-002A Measurements

For each candidate record:

$L(M)$

development predictive bits

sealed predictive bits

actual coded bits

T

calibration error

prediction interval coverage

development-to-test degradation

Repeat across hidden variation in:

D

x_0

trajectory length

sampling interval

seed

18 EXP-002A Preregistered Prediction

Expected behavior:

diffusion:
generalizes

diffusion + drift:
does not earn unnecessary drift parameter

mean reversion:
not systematically rewarded

curve fit:
may fit development trajectory
but fails on fresh trajectories

high-capacity predictor:
wins only if predictive savings exceed complexity

memorizer:
fails generalized gain

Gate:

EXP_002A_RANDOMNESS_DISCRIMINATION = PASS

19 EXP-002B: Occam Transition

19.1 Hidden Process

Generate:

$$dX_t = \mu dt + \sqrt{2D} dW_t.$$

The question is not merely whether drift can be detected. The question is when its predictive value becomes large enough to pay for the additional parameter.

20 Drift Sweep

Preregister a sweep:

mu = 0
very small
small
medium
large

Actual numeric values must be chosen through power simulation and frozen before final data generation. Where appropriate vary:

```
sign(mu)
D
x0
trajectory duration
```

21 Primary Comparison

Compare:

M0:
diffusion

M1:
diffusion + drift

Calculate $\Delta L(M)$, $\Delta L(D|M)$, and ΔT . At sufficiently small drift:

$$\Delta L(D|M) < \Delta L(M),$$

so drift should not earn its cost. At sufficiently strong drift:

$$\Delta L(D|M) > \Delta L(M),$$

so the richer explanation should win.

22 Occam Curve

Report:

```
effect size
delta model bits
delta predictive bits
delta T
uncertainty interval
probability richer model wins
```

Estimate the region where the richer model crosses from unjustified to justified. The $\mu = 0$ condition is mandatory. Systematic preference for drift at $\mu = 0$ is a failure requiring investigation.

Gate:

```
EXP_002B_OCCAM_TRANSITION = PASS
```

23 EXP-002C: Mechanism Discrimination

23.1 Hidden Process

Generate:

$$dX_t = -\theta X_t dt + \sigma dW_t.$$

This experiment tests a qualitatively different mechanism: state-dependent restoring dynamics.

24 Candidate Families

Compare:

Brownian motion
Brownian motion + constant drift
Ornstein-Uhlenbeck / mean reversion
flexible black-box predictor
trajectory-fitting model

25 Observational Phase

Develop candidates on a standard observational distribution. Freeze all candidate artifacts before intervention conditions are revealed.

26 Intervention Phase

Evaluate from:

near-equilibrium initial state

moderately displaced initial state

far-displaced initial state

A genuine mean-reverting explanation predicts that expected restoring behavior changes systematically with state. A trajectory-fitting predictor may fail.

27 Explanatory Evidence Rule

Do not call successful passive compression explanatory by itself. The stronger interpretation requires evidence such as:

invariance

interventional prediction

counterfactual success

mechanistic structure

A particularly strong result is:

model frozen
->
initial condition changed
->
new regime unseen during fitting
->

same compact law predicts outcomes
->
compression advantage survives

Gate:

EXP_002C_MECHANISM_DISCRIMINATION = PASS

28 EXP-002D: Unification

Construct a family containing:

pure diffusion
drifted diffusion
mean reversion
different D
different mu
different theta
different time steps
different initial states

Compare two architectures.

Specialists

Separate predictor for each environment. Charge every model independently.

Unified Theory

One common framework plus small environment-specific parameters. Charge:

shared theory
parameterization
environment parameters
selection metadata

The source explicitly proposes testing whether an explanation can compress multiple hidden worlds and earn aggregate gain through genuine reuse.

29 Unification Scaling Curve

Evaluate:

1 environment
2 environments
4 environments
8 environments
...

Observe whether the initial cost of a common abstraction eventually becomes cheaper than maintaining separate specialist theories. Also include a heterogeneous negative-control collection where forced unification should not win. Success requires both:

reward useful unification

reject artificial unification

Gate:

EXP_002D_UNIFICATION = PASS

30 EXP-003: Active Scientific Agency

This experiment asks whether learned explanations help determine what evidence should be collected next. It is potentially the most important step because it tests whether explanatory representations improve action rather than merely retrospective scoring.

31 Ambiguous Starting State

Construct observations compatible with at least two hypotheses. Example:

H0:

free diffusion

H1:

mean reversion

The initial evidence must not trivially distinguish them.

32 Permitted Experiments

Define a finite preregistered action space such as:

change initial displacement

increase observation duration

change sample frequency

request another replicate

Every permitted intervention receives an explicit cost.

33 Candidate Decision Loop

At every round:

current observations

->

competing hypotheses

->

predictive distribution under each possible intervention

->

expected information gain

->


```

rank interventions
->
commit chosen intervention
->
evaluator executes experiment
->
result revealed
->
hypotheses rescored

```

All predictions and choices must be committed before the result is revealed.

34 Passive Baselines

Freeze:

```

fixed observation schedule
random legal intervention
regular non-adaptive schedule

```

Use identical discrimination thresholds.

35 Primary EXP-003 Endpoint

Measure N_{active} against N_{passive} , where N is the number of observations required to cross a preregistered discrimination threshold. Also record:

```

time-to-discrimination
experimental operations
T accumulated
energy used

```

Before H4 passes, energy results must be labeled provisional.

Gate:

```
EXP_003_ACTIVE_EXPERIMENT_SELECTION = PASS
```

only if the active strategy uses meaningfully fewer observations or lower preregistered experimental cost than passive baselines.

36 Counterfactual Experiment Audit

After each completed active run, use the hidden simulator offline to calculate what would have happened under the other permitted interventions. The live candidate never receives these counterfactual results. This allows evaluation of whether its selected experiment was actually informative relative to alternatives.

37 H3: Reproducibility

Reproducibility contains two separate requirements.

37.1 H3-A: Replay

Using identical:

candidate
profile
data
seed

an independent implementation must reconstruct the same result within numerical tolerance.

37.2 H3-B: Statistical Robustness

Using:

new seeds
new trajectories

the exact Turing value may change. The preregistered scientific conclusion should remain.

Gate:

H3_REPLAY_REPRODUCIBILITY = PASS

H3_STATISTICAL_ROBUSTNESS = PASS

38 H4: Energy Attribution

T/J is not validated until the physical energy-accounting system itself is validated. Measure E_0 for idle/background operation, E_A for workload A, E_B for workload B, and E_{AB} for concurrent execution.

39 Interaction Energy

Calculate:

$$I_{AB} = E_{AB} - E_A - E_B + E_0.$$

Do not assume this value is zero. Concurrent workloads may exhibit:

contention
shared efficiency
thermal interaction
memory interaction
scheduler interaction

40 Energy Categories

Allow accounting to distinguish:

A-attributed
B-attributed
shared
interaction
unattributed
measurement uncertainty

Do not force every joule into one workload merely to make the accounting look clean.

41 H4 Test Matrix

Measure:

A alone
B alone
A+B

CPU-heavy + CPU-heavy

GPU-heavy + GPU-heavy

CPU-heavy + GPU-heavy

memory-bandwidth contention

light/heavy combinations

idle background

Randomize run order where practical. Repeat across representative thermal and load states.

42 Conservation Requirement

For each test require:

measured energy
~=
attributed
+ shared
+ interaction
+ unattributed

within preregistered sensor and integration uncertainty.

Gate:

H4_ENERGY_ACCOUNTING_CLOSURE = PASS

Until this passes, validated T/J claims are prohibited.

43 H5: Turing Yield

Once both T and energy are independently validated, compare semantically equivalent realizations.
Example:

reference CPU
optimized CPU
accelerated realization

The candidates must produce equivalent predictive distributions inside a preregistered tolerance.
Require:

$$T_A \approx T_B.$$

Then measure:

$$Y_A = \frac{T_A}{E_A}, \quad Y_B = \frac{T_B}{E_B}.$$

44 Separate Evaluation and Discovery Yield

Report:

Evaluation Yield

$$T/J_{\text{evaluation}}$$

Energy required to execute an already-discovered explanation.

Discovery Yield

$$T/J_{\text{discovery}}$$

Energy expended discovering that explanation. A system that requires enormous discovery energy but cheap inference must not be advertised solely by inference efficiency.

Gate:

H5_TURING_YIELD = PASS

requires at least one semantically equivalent pair whose T/J difference exceeds the combined uncertainty envelope.

45 Sensitivity Analysis

Every major successful result must be deliberately attacked. Vary reasonable preregistered alternatives for:

baseline
model-code scheme
parameter precision
probability quantization
reference coder
trajectory length
holdout regime

seed
uncertainty method
block-resampling assumptions

The numerical Turing value may change. The core question is whether scientific conclusions change.

46 Rank-Reversal Analysis

For every meaningful reversal classify the cause as:

decision-boundary effect
sample-size insufficiency
baseline sensitivity
model-code sensitivity
quantization sensitivity
statistical instability
implementation defect
fundamental metric sensitivity

Do not average reversals away.

Gate:

TURING_PROFILE_SENSITIVITY = PASS

only under a preregistered qualitative-stability criterion.

47 Independent Laboratory Replication

The framework should not be considered strongly validated until at least one external laboratory independently reproduces a subset of the program. The external laboratory should:

generate its own hidden observations

control its own blinding

implement the published measurement profile

write its own scorer

preferably write its own entropy coder

run EXP-001

run at least one full EXP-002 family

test one reasonable alternative profile

The central replication question is not:

Did both laboratories report exactly the same T?

It is:

Did the same scientific candidates
win and lose for the same reasons?

48 Experimental Artifact Standard

Each experiment should produce an immutable bundle. Recommended structure:

```
experiment/  
|-- preregistration.json  
|-- profile.json  
|-- profile.sha256  
|-- environment.json  
|-- candidate_manifest.json  
|-- candidates/  
|-- development_data/  
|-- sealed_data_commitment/  
|-- probability_streams/  
|-- coded_artifacts/  
|-- raw_measurements/  
|-- energy/  
|-- statistical_analysis/  
|-- independent_verification/  
|-- sensitivity/  
|-- replication/  
|-- final_receipt.json  
|-- REPORT.md
```

The sealed observations themselves may remain private until the declared release condition.

49 Canonical Experiment Receipt

Every experiment should emit a receipt containing:

```
experiment_id  
experiment_version  
  
profile_digest  
  
candidate_digests  
baseline_digest  
  
development_dataset_digest  
sealed_dataset_digest  
  
coder_digests  
scorer_digests
```

software_digest
hardware_identity

start_time
end_time

primary_endpoint
result

T
uncertainty

energy
energy_uncertainty

pass_fail

preregistered_gate_digest

artifact_root

verifier_signature

50 Statistical Analysis Principles

Independence

Do not treat correlated observations within a trajectory as independent samples. Use trajectory-level replication or block resampling as defined by the profile.

Multiple Comparisons

If many candidate families or parameters are evaluated, define the correction or hierarchical analysis before evaluation.

Failed Runs

A failed computation may be excluded only under preregistered technical-failure criteria. Scientifically inconvenient outcomes may not be discarded.

Stopping

No optional stopping unless explicitly modeled and preregistered.

Seeds

Seeds must be committed before final evaluation.

Uncertainty

Report uncertainty around:

T
delta T
model ordering
crossover point
T/J
energy attribution
active/passive efficiency

rather than publishing only point estimates.

51 Failure Policy

A failed gate should produce:

FAIL receipt
raw evidence
diagnosis
corrective hypothesis

The next version becomes a new experiment. Example:

EXP-002B-v1
FAIL

EXP-002B-v2
new preregistration
new sealed worlds

Do not silently rerun a failed experiment until success.

52 Claim Ladder

Scientific claims should be limited to the highest completed evidence level.

Level	Required Evidence	Permitted Interpretation
0	Definition only	mathematically specified statistic
1	EXP-001	reproducibly connected to actual lossless compression in tested regime
2	EXP-002A/B/C	discriminates supported from unsupported stochastic structure
3	interventions + EXP-002D	evidence for invariant/unifying explanatory structure
4	EXP-003	explanatory models improve experimental choice
5	H4	physical energy attribution validated
6	H5	explanatory efficiency measurable as T/J in tested regime
7	sensitivity	conclusion stable across reasonable profile choices
8	independent replication	independently reproducible metrology
9	multiple unrelated domains	evidence for broader applicability

53 Final Qualification Matrix

At the conclusion of the initial calibration program publish:

TURING_PROFILE_V1_FROZEN
PASS / FAIL

EXP_001_COMPRESSION_BRIDGE
PASS / FAIL

EXP_002A_RANDOMNESS_DISCRIMINATION
PASS / FAIL

EXP_002B_OCCAM_TRANSITION
PASS / FAIL

EXP_002C_MECHANISM_DISCRIMINATION
PASS / FAIL

EXP_002D_UNIFICATION
PASS / FAIL

EXP_003_ACTIVE_EXPERIMENT_SELECTION
PASS / FAIL

H3_REPLAY_REPRODUCIBILITY
PASS / FAIL

H3_STATISTICAL_ROBUSTNESS
PASS / FAIL

H4_ENERGY_ACCOUNTING_CLOSURE
PASS / FAIL

H5_TURING_YIELD
PASS / FAIL

TURING_PROFILE_SENSITIVITY
PASS / FAIL

INDEPENDENT_REPLICATION
PASS / FAIL

54 Recommended Execution Order

The engineering program should proceed in the following order:

1. Freeze model encoding and L(M)

2. Build probability-stream artifact format
3. Build independent arithmetic/range coder
4. Build independent ANS/rANS coder
5. Build independent Turing verifier
6. Complete CAL-0
7. Run EXP-001
8. Repeat EXP-001 on independent seeds
9. Run EXP-002A
10. Run EXP-002B
11. Run EXP-002C
12. Run interventional EXP-002C
13. Run EXP-002D
14. Run H3 internal replication
15. Run EXP-003
16. Validate physical energy measurement with H4
17. Run H5 Turing-yield experiments
18. Execute profile-sensitivity matrix
19. Freeze the complete calibration record
20. Hand the protocol to an independent laboratory

55 Immediate Milestone

The first milestone is deliberately narrow:

Produce one fully reproducible EXP-001 bundle in which frozen predictors emit immutable probability streams, two independently implemented lossless coders produce physical compressed artifacts, a separate verifier reconstructs all Turing values, and the explanatory candidate and memorization control behave according to the preregistered prediction.

Nothing later in the program is more important until that succeeds.

56 Final Scientific Standard

The validation program should not be judged by whether the experiments generate impressive Turing numbers. It should be judged by whether the instrument survives cases designed to falsify it.

A persuasive calibration record would show that the metric:

rewards correct diffusion

rejects invented drift when $\mu = 0$

rewards real drift only after it earns its description cost

rejects deterministic fits to stochastic fluctuations

recognizes mean reversion only when it exists

continues to predict after intervention

rewards useful theoretical unification

rejects artificial unification

helps choose discriminating experiments

reproduces under new trajectories

reproduces under independent implementation

survives reasonable profile changes

accounts physically for its energy use

distinguishes equivalent explanations by T/J

and reproduces in an independent laboratory

If the program achieves those outcomes under frozen, blinded, preregistered conditions, the Turing would have moved from a defined theoretical instrument to an empirically calibrated measurement framework. If it fails any of them, that failure identifies exactly which scientific interpretation has not yet been earned.